

VAPT INTERNSHIP

WEEK 6 Password Cracking & Steganography

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Task 1: Password Cracking

Objective: Crack the following MD5 hashes using any tool or online resource.

Tools Used: hashcat

Commands Used:

1. Step 1 - Save all hashes into a file
nano hashes.txt (paste the 5 hashes one per line, then save)
2. Step 2 - Identify hash type
hashid -m 5f4dcc3b5aa765d61d8327deb882cf99
3. Step 3 - Extract rockyou wordlist
sudo gunzip /usr/share/wordlists/rockyou.txt.gz
4. Step 4 - Run Hashcat dictionary attack (MD5 = mode 0)
hashcat -m 0 -a 0 hashes.txt /usr/share/wordlists/rockyou.txt -force
5. Step 5 - View cracked results
hashcat -m 0 hashes.txt -show

SN	Hash (MD5)	Plaintext Password	Notes
1	5f4dcc3b5aa765d61d8327deb882cf99	password	Most common password, cracked instantly
2	e10adc3949ba59abbe56e057f20f883e	123456	Top used password globally
3	25f9e794323b453885f5181f1b624d0b	123456789	Common numeric sequence
4	d8578edf8458ce06fbc5bb76a58c5ca4	qwerty	Keyboard pattern password
5	96e79218965eb72c92a549dd5a330112	111111	Repeated digits, cracked immediately

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Kali Lab - VMware Workstation
File Edit View VM Tabs Help
My Computer x Kali Lab x
kali@kali: ~/vaptweek6sulav
Session Actions Edit View Help
kali@kali: ~/vaptweek6sulav kali@kali: ~/vaptweek6sulav
(kali@kali)~$ nano hashes.txt
(kali@kali)~$ cat hashes.txt
5f4dcc3b5aa765d61d8327deb882cf99
e10adc3949ba59abbe56e057f20f883e
25f9e794323b453885f5181f1b624d0b
d8578edf8458ce06fbc5bb76a58c5ca4
96e79218965eb72c92a549dd5a330112
(kali@kali)~$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
[sudo] password for kali:
gzip: /usr/share/wordlists/rockyou.txt already exists; do you wish to overwrite (y or n)? y
(kali@kali)~$ hashcat -m 0 -a 0 hashes.txt /usr/share/wordlists/rockyou.txt --force
hashcat (v7.1.2) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEP, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i5-12500H, 2222/4445 MB (1024 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256
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kali@kali: ~/vaptweek6sulav kali@kali: ~/vaptweek6sulav
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c
Host memory allocated for this attack: 513 MB (3344 MB free)

Dictionary cache built:
* Filename ..: /usr/share/wordlists/rockyou.txt
* Passwords..: 14344392
* Bytes.....: 139921507
* Keyspace...: 14344385
* Runtime... : 0 secs

e10adc3949ba59abbe56e057f20f883e:123456
25f9e794323b453885f5181f1b624d0b:123456789
5f4dcc3b5aa765d61d8327deb882cf99:password
d8578edf8458ce06fbc5bb76a58c5ca4:qwerty
96e79218965eb72c92a549dd5a330112:111111

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 0 (MD5)
Hash.Target.....: hashes.txt
Time.Started....: Wed May 6 09:32:05 2026, (1 sec)
Time.Estimated...: Wed May 6 09:32:06 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 41444 H/s (1.00ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 5/5 (100.00%) Digests (total), 5/5 (100.00%) Digests (new)
Progress.....: 4096/14344385 (0.03%)
Rejected.....: 0/4096 (0.00%)
Restore.Point....: 0/14344385 (0.00%)
```

```
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kali@kali: ~/vaptweek6sulav kali@kali: ~/vaptweek6sulav

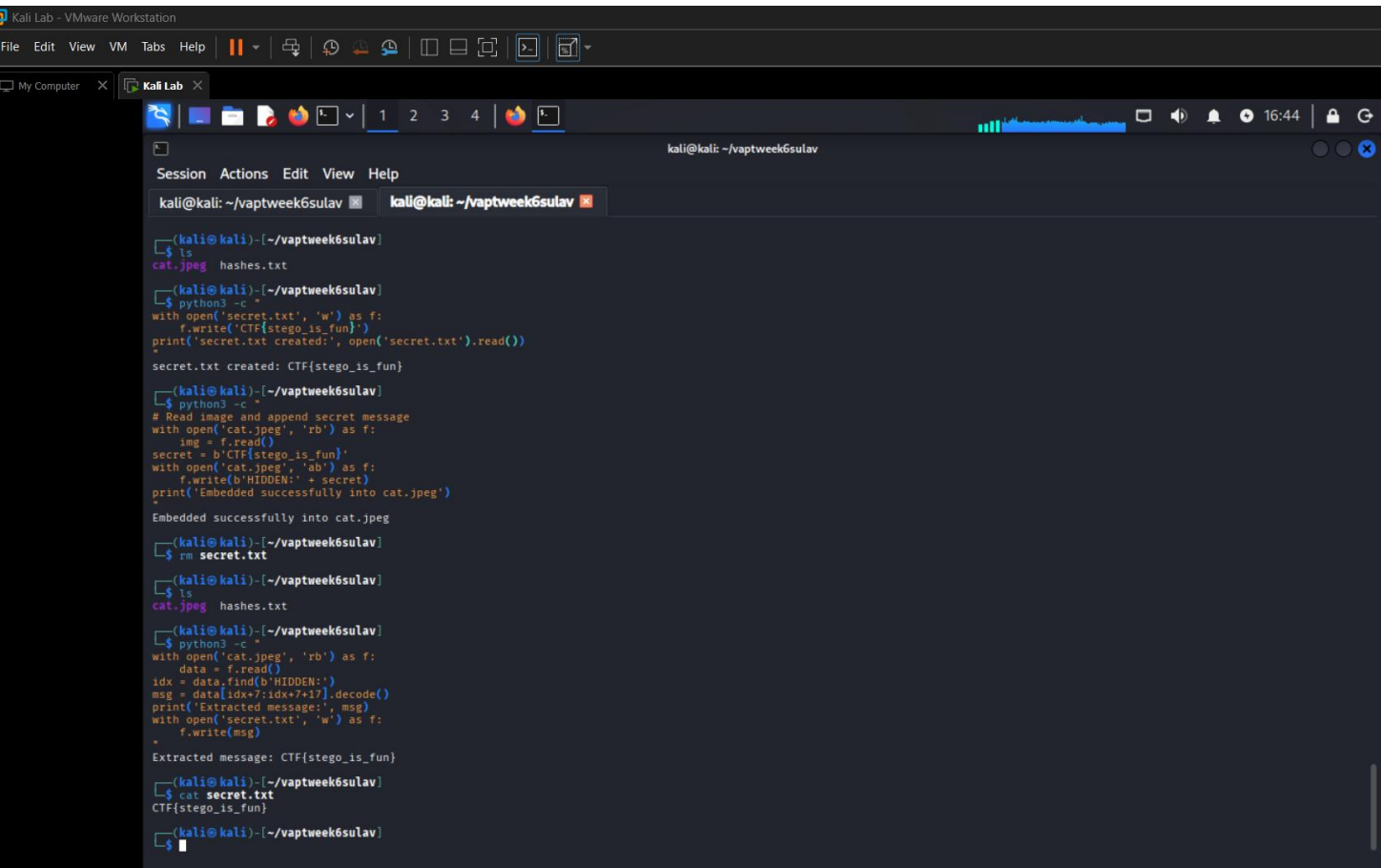
Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 0 (MD5)
Hash.Target.....: hashes.txt
Time.Started....: Wed May 6 09:32:05 2026, (1 sec)
Time.Estimated...: Wed May 6 09:32:06 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
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Recovered.....: 5/5 (100.00%) Digests (total), 5/5 (100.00%) Digests (new)
Progress.....: 4096/14344385 (0.03%)
Rejected.....: 0/4096 (0.00%)
Restore.Point....: 0/14344385 (0.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine..: Device Generator
Candidates.#01...: 123456 -> oooooo
Hardware.Mon.#01..: Util: 88%

Started: Wed May 6 09:31:31 2026
Stopped: Wed May 6 09:32:06 2026

(kali@kali)~$ hashcat -m 0 hashes.txt --show
5f4dcc3b5aa765d61d8327deb882cf99:password
e10adc3949ba59abbe56e057f20f883e:123456
25f9e794323b453885f5181f1b624d0b:123456789
d8578edf8458ce06fbc5bb76a58c5ca4:qwerty
96e79218965eb72c92a549dd5a330112:111111
(kali@kali)~$
```

Task 2: Steganography

Objective: Hide the secret message "CTF{stego_is_fun}" inside a JPEG image using steghide, then extract it. However, since steghide was not running on my Kali Linux machine, I used Python scripting instead.



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kali@kali:~/vaptweek6sulav
$ ls
cat.jpeg hashes.txt
$ python3 -c "
with open('secret.txt', 'w') as f:
    f.write('CTF{stego_is_fun}')
print('secret.txt created:', open('secret.txt').read())
"
secret.txt created: CTF{stego_is_fun}

kali@kali:~/vaptweek6sulav
$ python3 -c "
# Read image and append secret message
with open('cat.jpeg', 'rb') as f:
    img = f.read()
secret = b'CTF{stego_is_fun}'
with open('cat.jpeg', 'ab') as f:
    f.write(b'HIDDEN:' + secret)
print('Embedded successfully into cat.jpeg')
"
Embedded successfully into cat.jpeg

kali@kali:~/vaptweek6sulav
$ rm secret.txt

kali@kali:~/vaptweek6sulav
$ ls
cat.jpeg hashes.txt

kali@kali:~/vaptweek6sulav
$ python3 -c "
with open('cat.jpeg', 'rb') as f:
    data = f.read()
idx = data.find(b'HIDDEN:')
msg = data[idx+7:idx+7+17].decode()
print('Extracted message:', msg)
with open('secret.txt', 'w') as f:
    f.write(msg)
"
Extracted message: CTF{stego_is_fun}

kali@kali:~/vaptweek6sulav
$ cat secret.txt
CTF{stego_is_fun}

kali@kali:~/vaptweek6sulav
$
```